

# Awais Asghar

AI Engineer

+1 250 507 9045 | [aasghar.bee22seecs@seecs.edu.pk](mailto:aasghar.bee22seecs@seecs.edu.pk) | [LinkedIn](#) | [GitHub](#)

## INTERESTS

Machine Learning, Deep Learning, Computer Vision, Generative AI, LLMs, RAG, Data Science, Agentic AI

## EDUCATION

### National University of Sciences and Technology (NUST)

Islamabad, Pakistan

*Bachelor of Engineering in Electrical Engineering*

*Sept 2022 - June 2026*

- CGPA: 3.63/4.0    Specialization GPA: 3.94/4.0    Merit Scholarship, 2024 - 2026
- Senior Design Thesis: ML-Based Hardware Accelerator for Real-Time Image Segmentation on FPGA

## RESEARCH EXPERIENCE

### Mitacs Globalink Research Intern

Summer 2026

*NRC Herzberg Institute of Astrophysics*

*Victoria, BC, Canada*

- Conduct applied ML and AI research on large-scale astrophysical datasets, contributing to model development, scientific computing workflows, and data analysis pipelines under Dr. Sebastien Fabbro
- Carrying out end-to-end academic research: literature review, experiment design, result analysis, and scientific documentation

### Deep Learning Research Intern

June 2024 - Sept 2024

*Deep Learning Lab, SINES, NUST*

*Islamabad, Pakistan*

- Built and evaluated CNN and transfer learning models in PyTorch and TensorFlow with GPU-accelerated training via CUDA, focused on performance benchmarking and model validation

### Machine Learning Research Intern

June 2024 - Aug 2024

*HamsanTech, NSTP*

*Islamabad, Pakistan*

- Built a multi-model skin cancer classification pipeline achieving 91% accuracy and 0.963 AUC using an ensemble of XGBoost, AdaBoost, LightGBM, SVM, and Logistic Regression
- Developed interactive Power BI and Tableau dashboards and ran Excel-based data cleaning and exploratory analysis on real industry datasets

## SELECTED PROJECTS

### ML-Based Hardware Accelerator for Image Segmentation | *Python, PyTorch, Vitis HLS, Vivado*

2025 - 2026

- Achieved 56x model compression to 570 KB with 97% pixel accuracy at 2.72W power on a Zybo Z7-20 FPGA, covering the full pipeline from training and quantization to edge deployment

### AI-Based Exam Generator (RAG) | *Python, LangChain, Groq API, Llama 3.1* | [Github Repo](#)

2025

- Built an end-to-end RAG pipeline combining document ingestion, vector embeddings, semantic retrieval, and LLM-based question generation into a production-ready system

### Lightweight U-Net for Driving Scene Segmentation | *Python, PyTorch, OpenCV* | [Github Repo](#)

Dec 2025

- Compressed a U-Net from 8.7M to 0.53M parameters while holding 72% IoU, 80% F1, and 96% test accuracy on the CARLA simulator dataset; paper in progress

### Real-Time Fabric Defect Detection | *Python, OpenCV, NumPy, Jetson Nano* | [Github Repo](#)

Dec 2025

- Built a classical computer vision pipeline (GLCM, FFT, Gabor filters, IoU-based fusion) for interpretable, low-power defect localization on edge hardware, with no deep learning required; paper in progress

### RecycleVision: Automated Waste Classification | *TensorFlow, Keras, EfficientNetB0* | [Github Repo](#)

Nov 2025

- Trained an EfficientNetB0 transfer learning model reaching 92.9% test accuracy and a 0.93 macro F1 score across multiple waste categories

### SkinSense: Multi-Model Skin Cancer Classifier | *Scikit-learn, XGBoost, PyTorch* | [Github Repo](#)

2024

- Built an ensemble classification pipeline achieving 91% accuracy and 0.963 AUC for single-image RGB skin cancer detection

## TECHNICAL SKILLS

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**Machine Learning & AI:** Python, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, SciPy, CUDA  
**LLMs & RAG:** LangChain, Groq API, Llama 3.1, Vector Databases, Retrieval-Augmented Generation, Prompt Engineering  
**Data Analysis & Visualization:** Excel, Power BI, Tableau, Matplotlib, Seaborn  
**Tools & Platforms:** Git/GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Kaggle, LaTeX

## PUBLICATIONS (IN PROGRESS)

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**Design of a Novel Lightweight U-Net Architecture for Efficient Semantic Segmentation:** based on FYP and autonomous driving segmentation research at NUST SEECS  
**Real-Time Fabric Defect Detection on Edge Devices Using Classical Computer Vision:** multi-method pipeline deployed on Jetson Nano

## CERTIFICATIONS AND RELEVANT COURSEWORK

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**Machine Learning Specialization** (3-Course Series), deeplearning.ai: supervised ML, advanced learning algorithms, unsupervised learning, reinforcement learning  
**Deep Learning Specialization** (5-Course Series), deeplearning.ai: neural networks, hyperparameter tuning, CNNs, RNNs, LSTMs, sequence modeling, NLP  
**Deep Learning Course Series**, MIT 6.S191: computer vision, generative modeling, transformers, LLMs, deep reinforcement learning, AI in biology  
**NUST Coursework:** Machine Learning, Deep Learning, Computer Vision, Linear Algebra, Calculus, Digital Signal Processing, Signals and Systems

## HONORS AND ACHIEVEMENTS

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- Mitacs Globalink Research Internship: fully funded international placement at NRC Herzberg, Canada (Summer 2026)
- Millennium Fellow: selected among the top 4% globally by the UN Academic Impact and Millennium Campus Network
- NUST SEECS Merit Scholarship, 2024 - 2026
- Dean's Honor List: recognized 4 times for outstanding academic performance
- Prime Minister Youth Laptop Scheme, 2025, awarded for academic excellence
- 3rd place, BISE Multan Pre-Engineering, among 70,000+ students
- Intermediate Scholarship: 100% fee waiver based on exceptional academic performance